IEOR 221

Homework 11

Module 11-12 Portfolio Management & Financial Engineering (November 19, 2025)

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1) Suppose that the market can be described by the following three sources of systematic risk with associated risk premiums.

Factor	Risk Premium
Industrial production (I)	6%
Interest rate (R)	2%
Consumer confidence (C)	4%

The return on a particular stock is generated according to the following equation:

$$r = 15\% + 1.0I + 0.5R + 0.75C + e$$

Find the equilibrium rate of return on this stock using the APT. The risk free rate is 6%. Is the stock over- or under-priced? Explain.

2) A trader finds shares of company A and B are out of their expected relative pricing pattern. Shares of A are under-priced while shares of B are over priced. The trader decided to use futures on A and B to set up a pair trading position:

$$\Pi = N_A V_A - N_B V_B + N_I V_I$$

N_A, N_B, N_I are the number of future contracts on A, B and market index, V_A, V_B, V_I are the value of the future contracts. N_I is chosen to hedge out the residual beta of the long-short position so that Π is market neutral. Currently $V_A = 100$, $V_B = 50$, $V_I = 1250$. The trader longs 1000 future contracts on A while shorting 2000 futures on B. $\beta_A = 0.8$, $\beta_B = 1.1$. What should the trader's position in N_I be?